

22/08/25
EXP:-

5. Study of Activation Functions and its role

Aim:-

To study different activation function and understand their role.

Objective:-

1. To implement and visualize common activation functions such as sigmoid, Tanh, ReLU and softmax.
2. To understand the importance of activation function in transforming input symbols.
3. To observe how activation functions affect the training & performs of models.

Pseudocode:-

Start.

1. import necessary libraries (numpy, matplotlib, etc.)
2. Define activation function.

$$(i) \text{ sigmoid}(x) = 1 / (1 + e^{(-x)})$$

$$(ii) \text{ tanh}(x) = (e^x - e^{(-x)}) / (e^x + e^{(-x)})$$

$$(iii) \text{ ReLU}(x) = \max(0, x)$$

$$(iv) \text{ softmax}(x_i) = e^{x_i} / \sum e^{x_j} \text{ for all } j$$

3. Generate input value in a range (eg:- -10 to 10)
4. Apply each activation function on the input values.
5. plot the output of each function to visualize their behaviour.
6. compare and observe.
 - range of outputs

- non-linearity introduced.
- suitability for classification/regression.

End.

Observation:-

1. Sigmoid: squashes values between 0 and 1. useful for probability but suffers from vanishing gradients.
2. Tanh: Squashes value between -1 and 1. centred at zero better than sigmoid in some cases.
3. ReLU: Output 0 for negative input and linear for positive. avoids vanishing gradients commonly used in hidden layer.
4. Softmax: Convert output into probabilities that sum to 1. mainly used in final layer of classification tasks.

Result:-

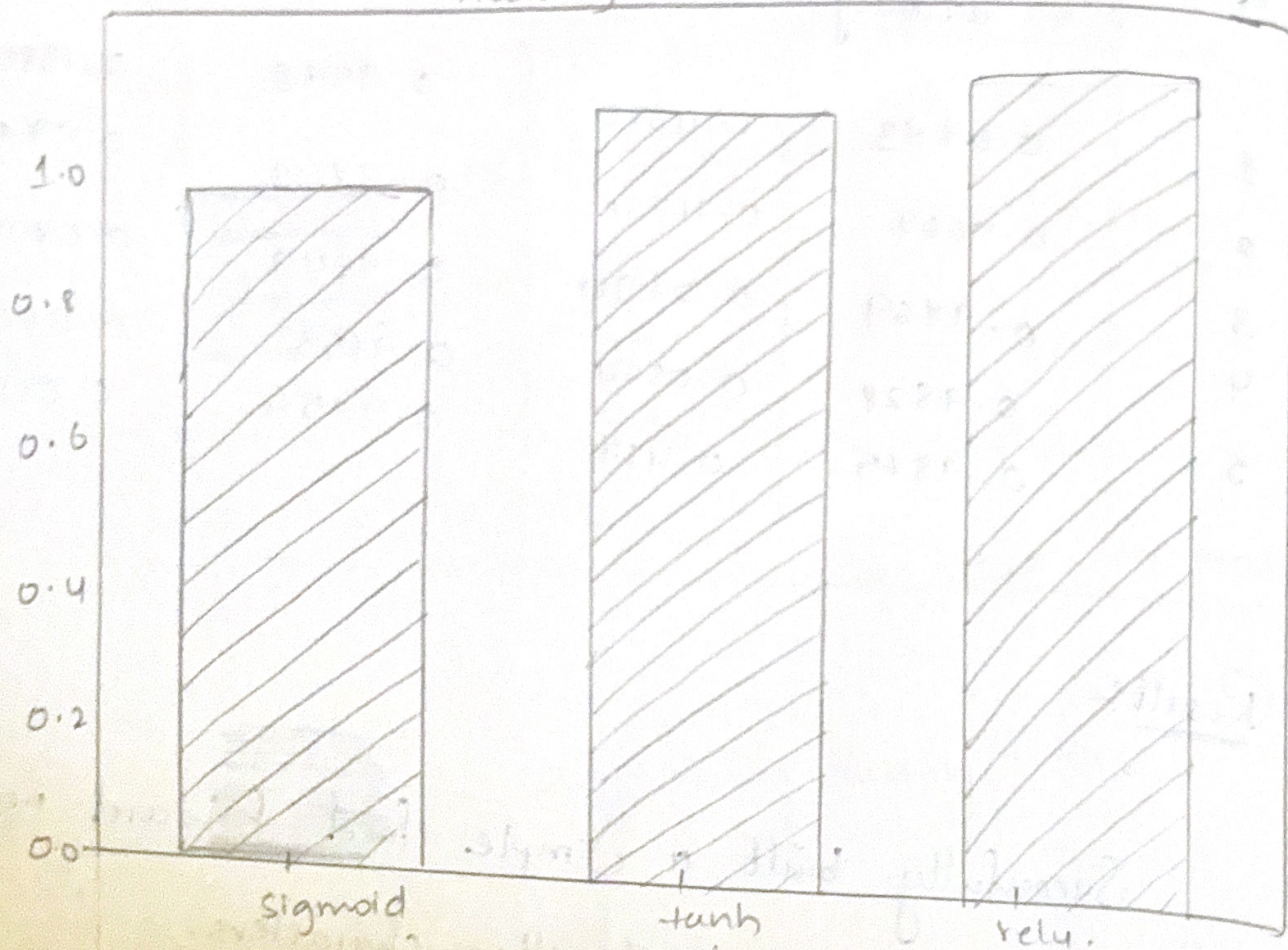
Successfully studied and implement activation functions and understood their role.

Training with Sigmoid. activation

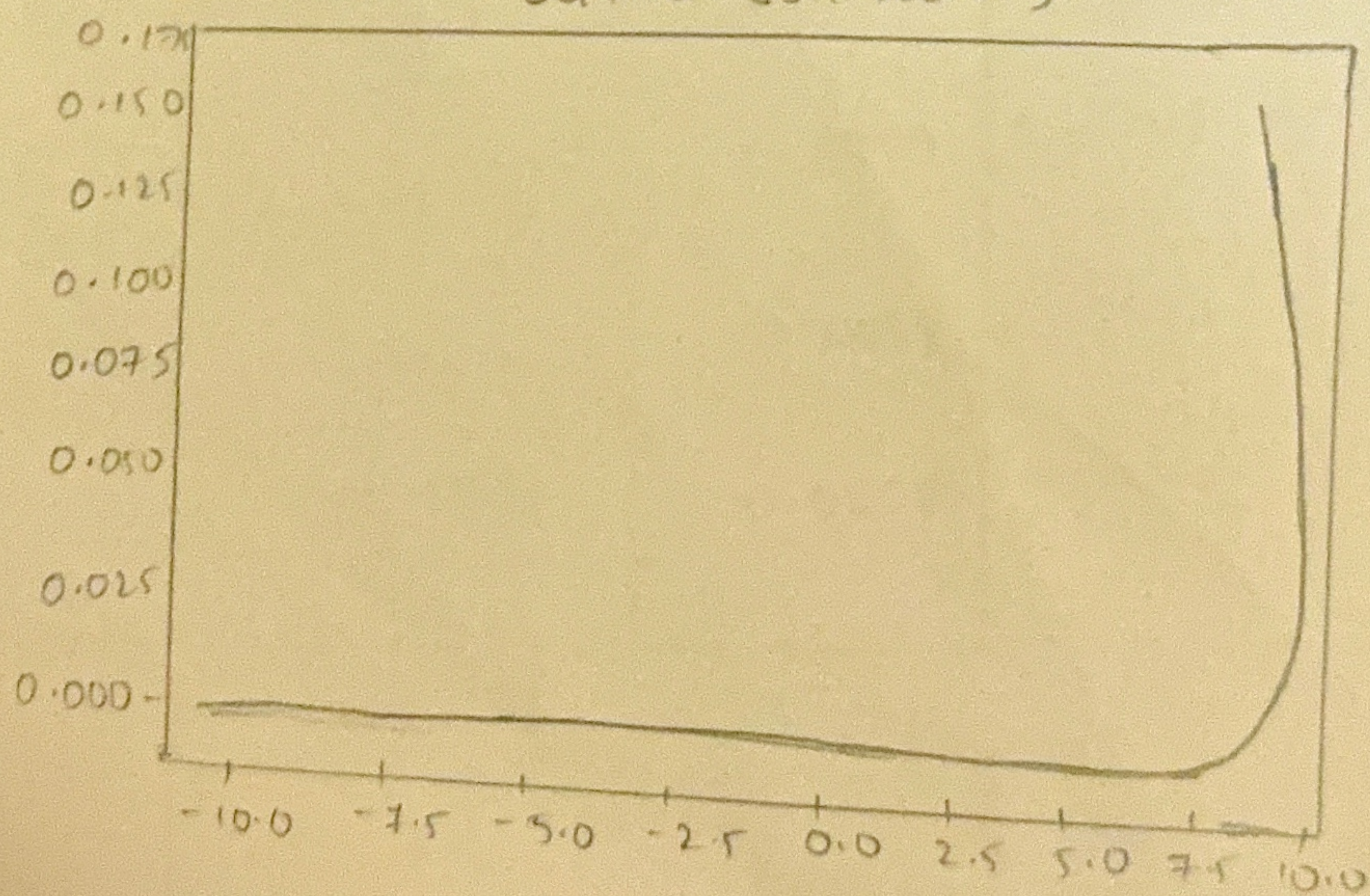
Training with tanh activation

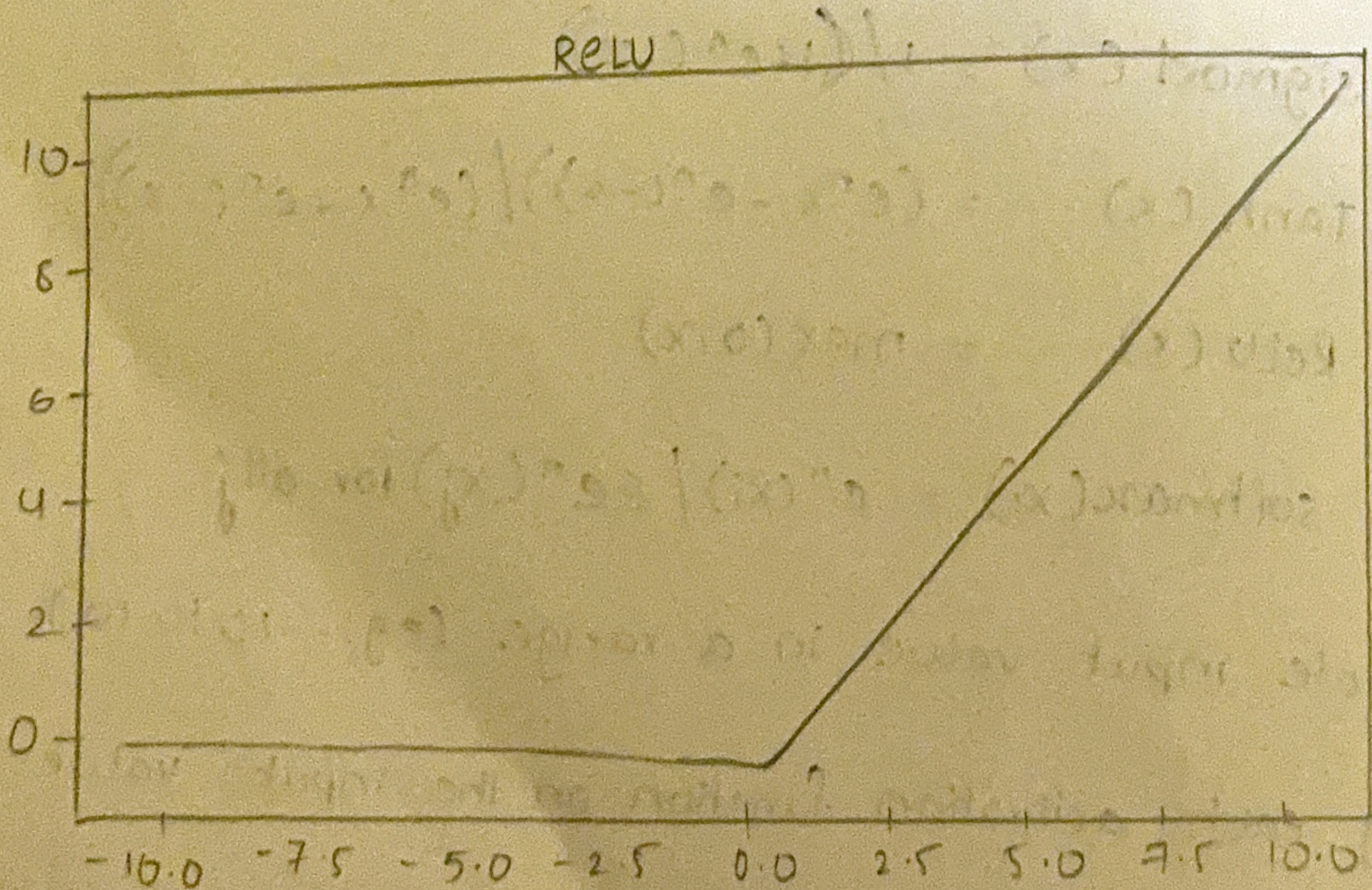
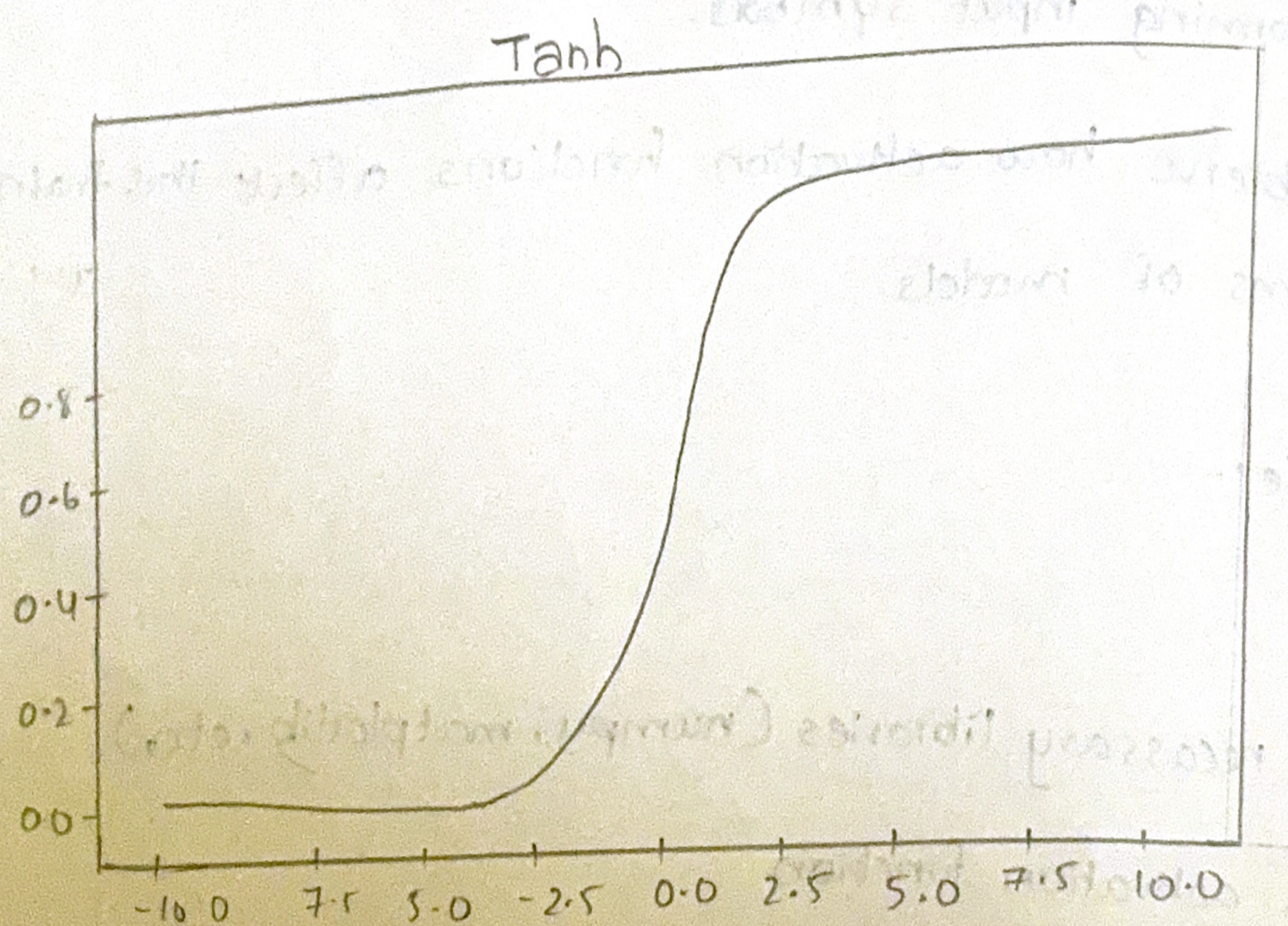
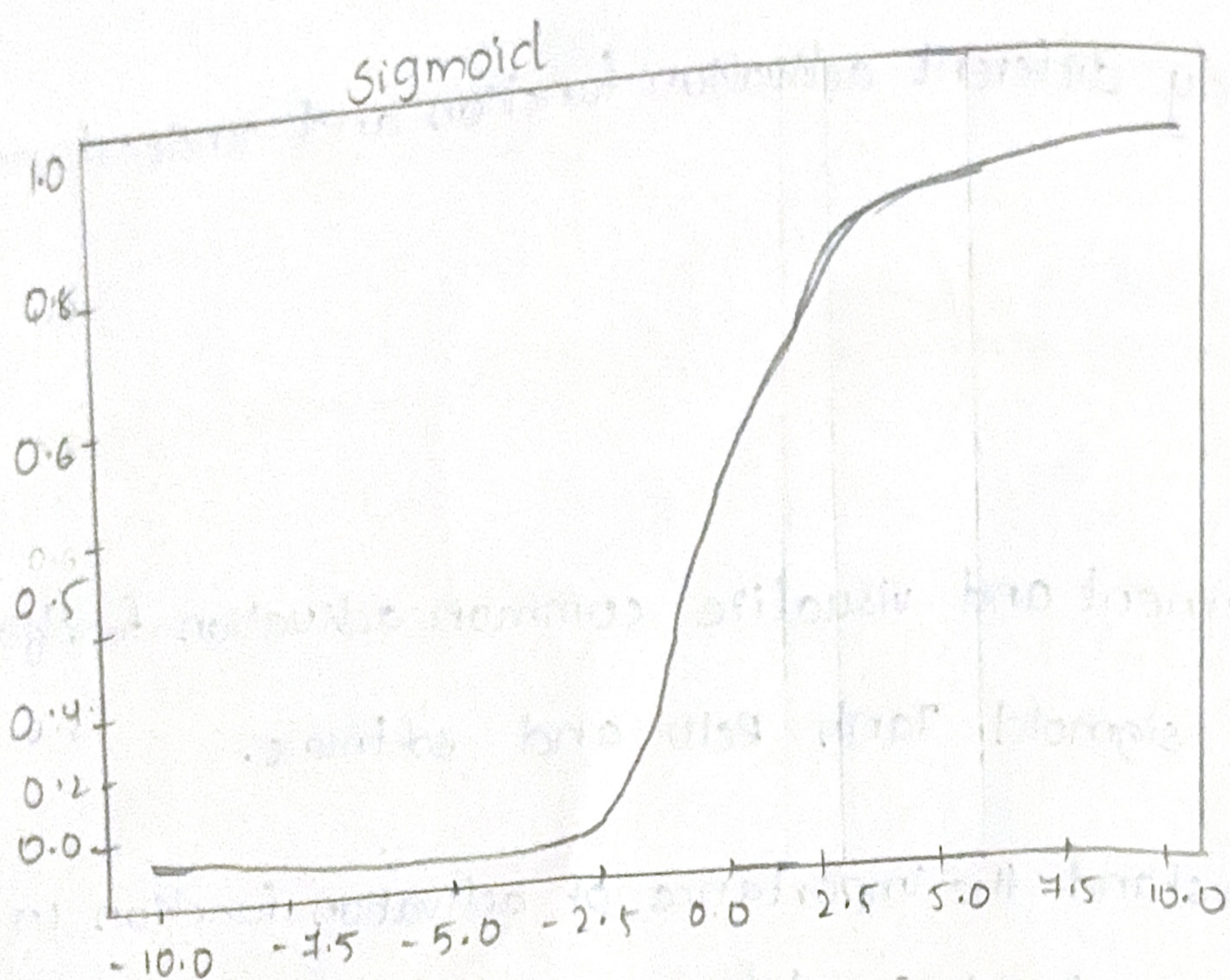
Training with relu activation.

Accuracy with different Activation function



softmax (on vector)





final results:-

Sigmoid : 93.69%

tanh : 96.18%

relu : 96.00%